University of California, Riverside
Electrical and Computer Engineering Department

Electromagnetics I

Lecture 9: Summary and Preparation for the Midterm Exam

April 30, 2026

Instructor: Dr. Alexander Khitun
Electrical and Computer Engineering Department, WCH 349
Email: akhitun@ece.ucr.edu

Office Hours: Thursday, 12:00 - 1:00 PM

Course Timeline

03/30/2026 - 06/05/2026

2:00 PM - 3:20 PM

APRIL 2026

SUN	MON	TUE	WED	THU	FRI	SAT
29	30	31	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	1	2

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MAY 2026

SUN	MON	TUE	WED	THU	FRI	SAT
6	27	28	29	30	1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31	1	2	3	4	5	6

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No-fail test

Midterm Exam

Summary and Preparation for the Exam

Announcement

- Midterm will be on May 5th

It is a closed-book exam. No notebooks or phones are allowed.

You may use one sheet of paper, format A4 (both sides) for notes.

Please bring your student ID

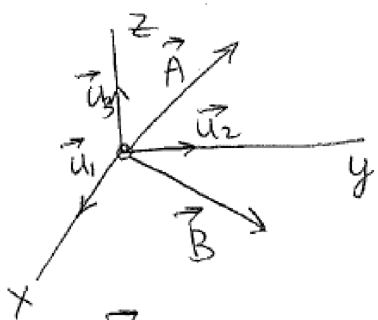
Midterm Exam

- 10 Multiple choice questions (2 points each)
- 7 problems (5, 10, 10, 10, 15, 10, and 20 points)
- 100 points total = 100%
- Extra points = Test -1 score/10
- 100 is the max you can get on the Midterm

VECTOR REPRESENTATION IN THE COORDINATE SYSTEMS

The vector $\vec{A}$ can be represented in terms of its components

$$\vec{A} = A_x \cdot \vec{a_x} + A_y \vec{a_y} + A_z \vec{a_z}$$



$$\vec{A} = A_1 \vec{u_1} + A_2 \vec{u_2} + A_3 \vec{u_3}$$

$$\vec{B} = B_1 \vec{u_1} + B_2 \vec{u_2} + B_3 \vec{u_3}$$

$$\vec{A} + \vec{B} = (A_1 + B_1)\vec{u_1} + (A_2 + B_2)\vec{u_2} + (A_3 + B_3)\vec{u_3}$$

Properties of unit vectors:

$$\vec{u_1} \cdot \vec{u_2} = 0$$

$$\vec{u_1} \cdot \vec{u_1} = 1$$

$$\vec{A} \cdot \vec{B} = (A_1 \vec{u_1} + A_2 \vec{u_2} + A_3 \vec{u_3}) \cdot (B_1 \vec{u_1} + B_2 \vec{u_2} + B_3 \vec{u_3}) =$$

$$= A_1 \cdot B_1 + A_2 \cdot B_2 + A_3 \cdot B_3$$

The cross product

$$\vec{A} \times \vec{B} = \begin{vmatrix} \vec{u}_1 & \vec{u}_2 & \vec{u}_3 \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix} =$$

$$= \vec{u}_1 \begin{vmatrix} A_2 & A_3 \\ B_2 & B_3 \end{vmatrix} - \vec{u}_2 \begin{vmatrix} A_1 & A_3 \\ B_1 & B_3 \end{vmatrix} + \vec{u}_3 \begin{vmatrix} A_1 & A_2 \\ B_1 & B_2 \end{vmatrix} =$$

$$= \vec{u}_1 \left(A_2 \cdot B_3 - A_3 B_2\right) - \vec{u}_2 \left(A_1 \cdot B_3 - A_3 B_1\right) + \vec{u}_3 \left(A_1 B_2 - A_2 B_1\right)$$

$$= \vec{u}_1 \left(A_2 B_3 - A_3 B_2\right) + \vec{u}_2 \left(A_3 B_1 - A_1 B_3\right) + \vec{u}_3 \left(A_1 B_2 - A_2 B_1\right)$$

This follows from the properties

$$\vec{u}_1 \times \vec{u}_2 = \vec{u}_3 \quad , \quad \vec{u}_2 \times \vec{u}_3 = \vec{u}_1 \quad , \quad \vec{u}_3 \times \vec{u}_1 = \vec{u}_2$$

$$\vec{u}_1 \times \vec{u}_1 = \vec{u}_2 \times \vec{u}_2 = \vec{u}_3 \times \vec{u}_3 = 0$$

Chapter 2: Maxwell's Equations in Differential Form

INTEGRAL FORMS	DIFFERENTIAL FORMS	
$\oint_s \epsilon_o \epsilon_r \mathbf{E} \cdot d\mathbf{s} = \int_v \rho_v \, dv$	$\nabla \cdot \epsilon_o \epsilon_r \mathbf{E} = \rho_v$	(4.1)
$\oint_s \mathbf{B} \cdot d\mathbf{s} = 0$	$\nabla \cdot \mathbf{B} = 0$	(4.2)
$\oint_c \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt}\int_s \mathbf{B} \cdot d\mathbf{s}$	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	(4.3)
$\oint_c \mathbf{H} \cdot d\boldsymbol{\ell} = \int_s \mathbf{J} \cdot d\mathbf{s} + \frac{d}{dt}\int_s \epsilon_o \epsilon_r \mathbf{E} \cdot d\mathbf{s}$	$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \epsilon_o \epsilon_r \mathbf{E}}{\partial t}$	(4.4)

TABLE B.1 NAMES AND SI UNITS OF QUANTITIES COMMONLY USED IN ELECTROMAGNETICS (Continued)

Quantity	Description	Units
Charge density (ρ)	either linear charge density ρ_l(= Q/length), surface charge density ρ_s (= Q/area) or volume charge density ρ_v (= Q/volume)	ρ_l in $\frac{\text{Coulomb}}{\text{meter}} = \frac{\text{C}}{\text{m}}$ ρ_s in $\frac{\text{Coulomb}}{\text{meter}^2} = \frac{\text{C}}{\text{m}^2}$ ρ_v in $\frac{\text{Coulomb}}{\text{meter}^3} = \frac{\text{C}}{\text{m}^3}$
Charge (q, Q)	current × time	Coulomb (C)
Conductance	1/resistance	Siemens (S) or Ω^{-1}
Conductivity (σ)	conductance/length	Siemens/meter (S/m)
Current (I)	charge/time	Ampere (A)
Current density	surface sheet current density $\mathbf{J}_s$ volume current density $\mathbf{J}$	Ampere/meter (A/m) Ampere/meter^2 (A/m^2)
Electric dipole (**p**)	$\mathbf{p} = q\mathbf{d}$, charge × distance	Coulomb × meter = C · m
Electric field intensity (**E**)	force/charge or potential/length	Volt/meter
Electric flux (ψ_e)	$\psi_e = \oint_s \mathbf{D} \cdot d\mathbf{s} = Q$	Coulomb (C)
Electric flux density (**D**)	electric flux/area	Coulomb/meter^2 (C/m^2)
Electric susceptibility (χ_e)	polarization $\mathbf{P} = \epsilon_o \chi_e \mathbf{E}$	χ_e is dimensionless
Force (**F**)	mass × acceleration	Newton (N)
Frequency (f)	cycles/second	hertz (Hz) (s^{-1})
Impedance (Z_o, η)	characteristic impedance of transmission line Z_o intrinsic impedance of free-space vacuum η_o	ohms (Ω) ohms (Ω)
Inductance (L)	magnetic flux/current	Wb/A or Henry (H)
Length (ℓ,d)		meter (m)
Magnetic dipole (**m**)	$\mathbf{m} = I\,d\mathbf{s}$, current × area	Ampere × meter^2 = A · m^2

TABLE B.1 NAMES AND SI UNITS OF QUANTITIES COMMONLY USED IN ELECTROMAGNETICS (Continued)

Quantity	Description	Units
Magnetic field intensity (**H**)	$\oint_c \mathbf{H} \cdot d\boldsymbol{\ell} = \text{I}$ or **H** is $\frac{\text{current}}{\text{distance}}$	Ampere/meter (A/m)
Magnetic flux (ψ_m)	$\psi_m = \int_s \mathbf{B} \cdot d\mathbf{s}$	Weber (Wb)
Magnetic flux density (**B**)	magnetic flux/area	Weber/m^2 (Wb/m^2)
Magnetic flux linkage	flux × number of turns	Weber · turn (Wb · turn)
Magnetic susceptibility (χ_m)	magnetization $\mathbf{M} = \chi_m \mathbf{H}$	χ_m is dimensionless
Magnetic vector potential (**A**)	$\mathbf{B} = \nabla \times \mathbf{A}$	Weber/meter (Wb/m)
Magnetization (**M**)	**M** ≡ magnetic dipole/volume	$\frac{\text{A} \cdot \text{m}^2}{\text{m}^3} = \text{A/m}$
Mass (m)		kilogram (kg)
Period (T)	1/frequency	second (s)
Permeability (μ_o)	for vacuum	Henry/meter (H/m)
Permittivity (ϵ_o) (dielectric constant)	for vacuum or air at atmospheric pressure	Farad/meter (F/m)
Phase constant (β)	change in phase of wave per unit length	radian/meter (rad/m)
Polarization (**P**)	electric dipole/volume	Coulomb/meter^2 = C/m^2
Potential Φ or voltage (V)	work/charge	Volt (V)
Power	energy/time	Watt (W)
Power density (**P**) vector (Poynting vector)	power/area	Watt/meter^2 (W/m^2)
Propagation constant ($\hat{\gamma}$)	$e^{\pm\hat{\gamma}z}$ for a wave propagating along ($\pm z$) axis	meter^{-1} (m^{-1})
Radian frequency (ω)	2π × frequency	radian/second (rad/s)
Reactance	voltage/current	ohms (Ω)

TABLE D.1 FREE-SPACE (VACUUM) CONSTANTS

Quantity	Symbol	Value
Velocity of light	c	2.997925×10^8 (m/s) $\sim 3 \times 10^8$ (m/s)
Dielectric constant (permittivity)	ϵ_o	8.854×10^{-12} (F/m) $\approx \frac{1}{36\pi} \times 10^{-9}$ (F/m)
Permeability	μ_o	$4\pi \times 10^{-7}$ (H/m)
Intrinsic impedance	$\eta_o = \sqrt{\frac{\mu_o}{\epsilon_o}}$	$120\pi \sim 377\ \Omega$
Wavelength λ_o	c/f (f is frequency)	meter (m)

Gauss's law for electric field

$$\boxed{\epsilon_0 \oint_S \vec{E} \cdot d\vec{S} = \int_V \rho_v \, dV}$$

closed surface integral

volume integral

Gauss' Law for Magnetic Field

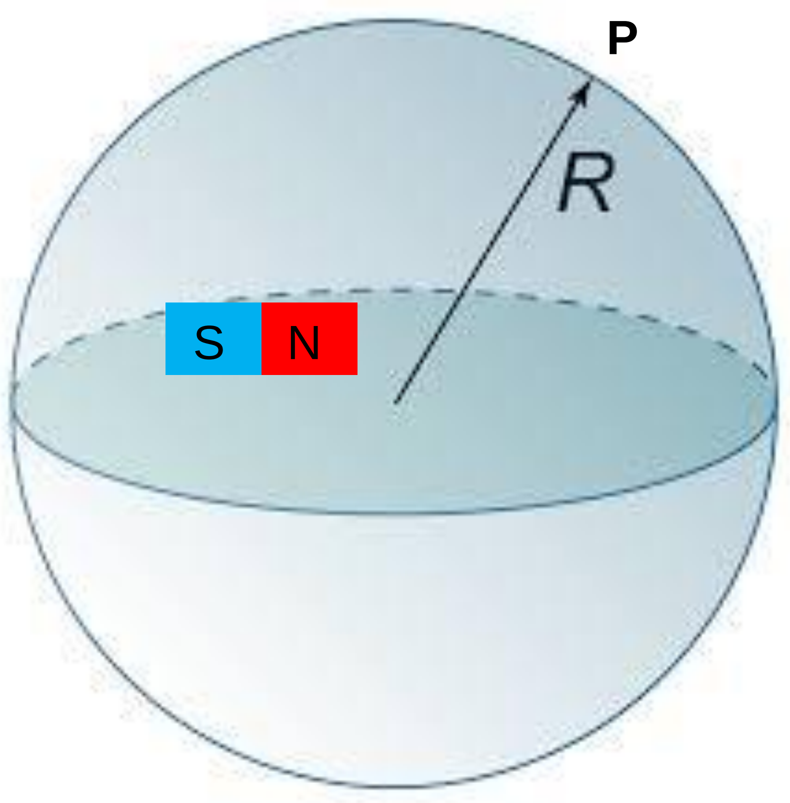
closed surface integral

$$\boxed{\oint_S \vec{B}\cdot d\vec{S} = 0}$$

Total magnetic flux through the closed surface should be equal to zero

It does not mean that the magnetic field is zero everywhere!

Retrieval Session



$div\vec{B} = 0$
$\vec{B} \neq 0$

Divergence at point P: 0 or Not 0
Magnetic Field at point P: 0, or Not 0, or N/A

Faraday's law

Differential form

Reminder that del is a vector operator

Reminder that the electric field is a vector

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

The rate of change of the magnetic induction with time

The differential operator called "del" or "nabla"

The cross-product turns the del operator into the curl

The electric field in V/m

Ampere's Law

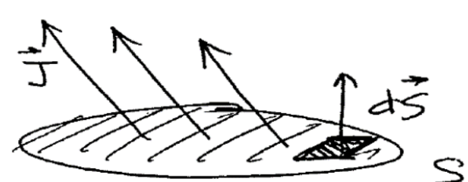
> The line integral of the magnetic flux density along a closed contour C is equal to the total current crossing the area S enclosed by this contour.

$$\oint_C \frac{\vec{B}}{\mu_0} \cdot d\vec{\ell} = \int_S \vec{J} \cdot d\vec{S} + \frac{d}{dt} \int_S \epsilon_0 \vec{E} \cdot d\vec{S}$$

$\int_S \vec{J} \cdot d\vec{S}$ – current resulting from flow of charges

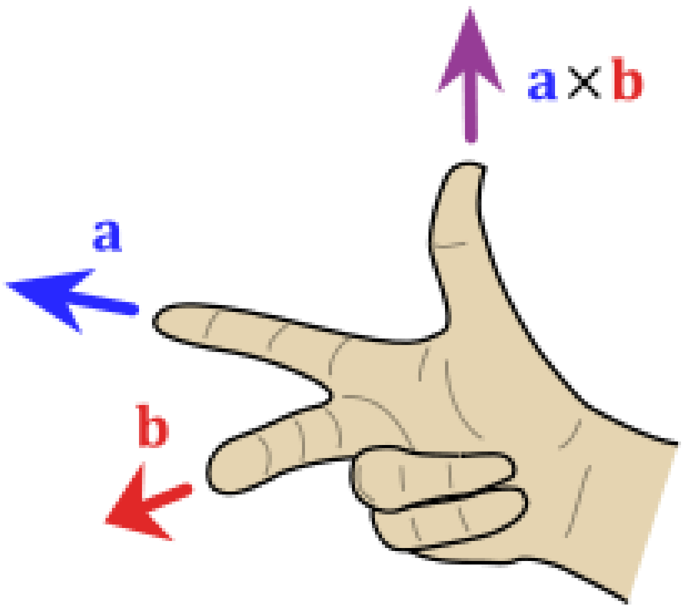
$\vec{J}$ – current flux density

$\vec{J} = \rho \cdot \vec{V}$

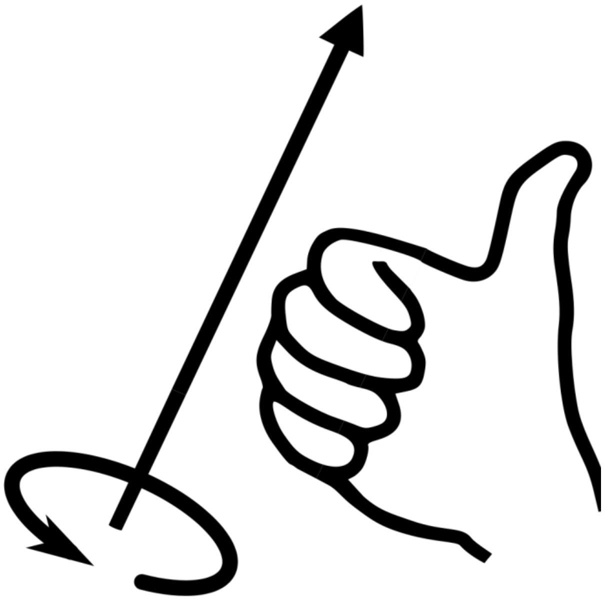


$\frac{d}{dt} \int_S \epsilon_0 \vec{E} \cdot d\vec{S}$ – "displacement" current (added by Maxwell)

Right-hand Rule



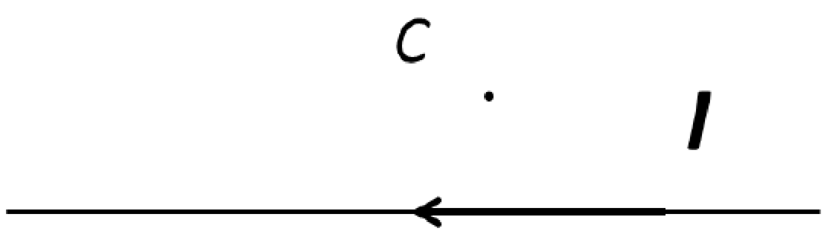
Finding the direction of the cross product by the right-hand rule



Right hand rule for curve orientation.

2. (**3 points**) The dot product of two non-zero vectors **E** and **B,** which characterize two vector fields, is zero. This means that these vectors are (circle the correct answer)
   a) not defined
   b) perpendicular to each other
   c) parallel to each other

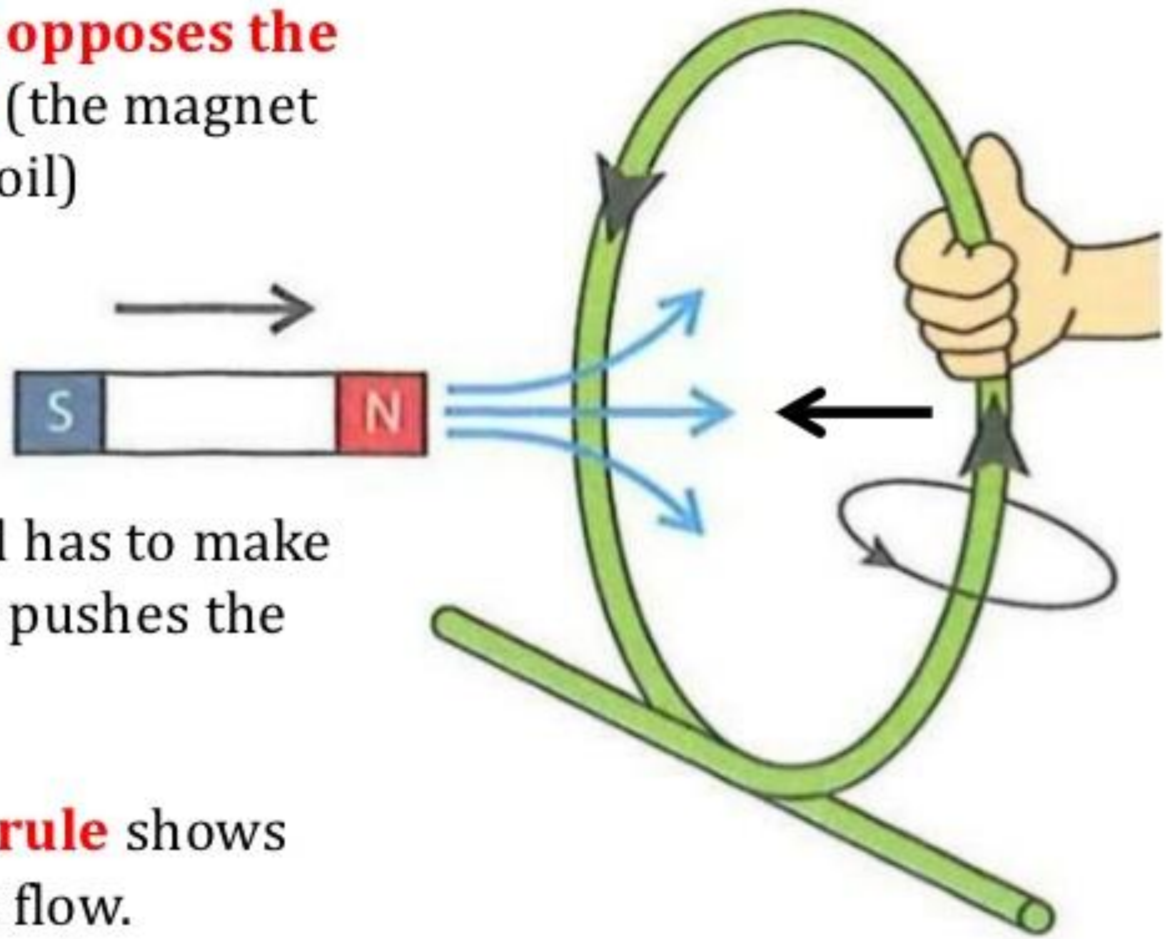
3. (**3 points**) There is a thin straight conductor carrying electric current *I* (see figure). What is the direction of the magnetic field in point C?
   a) up
   b) down
   c) to us
   d) from us



Faraday's and Lenz's Laws – Example 1

B) Lenz's Law

The direction of the Induced current is such that **it opposes the change producing it** (the magnet moving towards the coil)

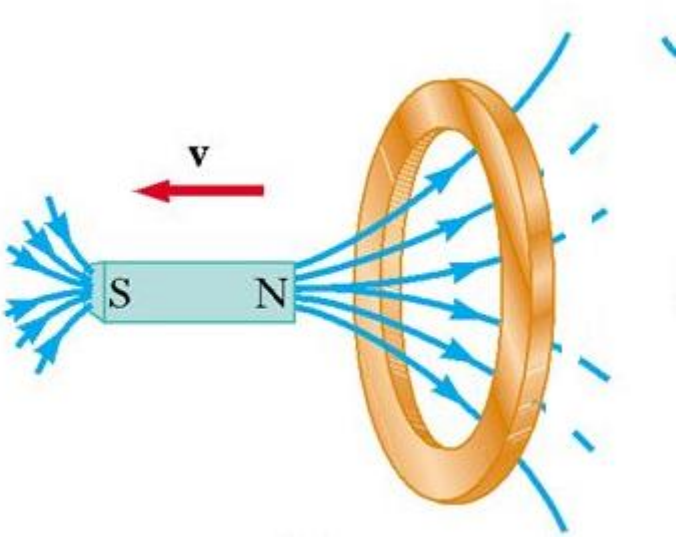


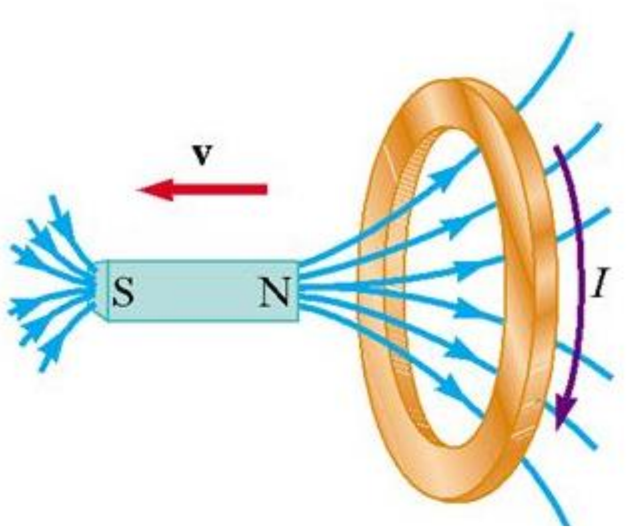
The current in the coil has to make a **magnetic field** that pushes the Magnet away....

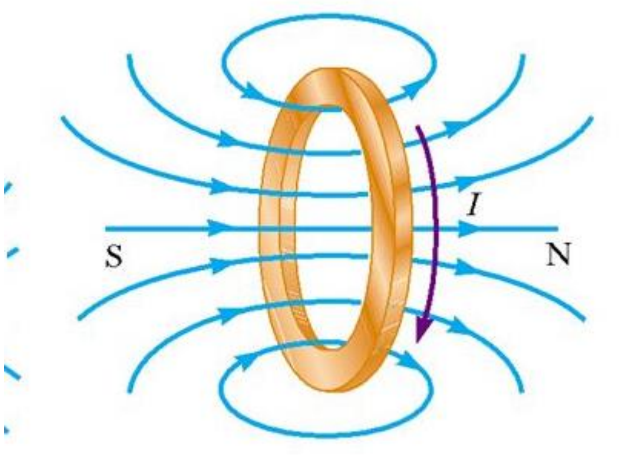
The **Right hand grip rule** shows how the current must flow.

Retrieval Session

What is the direction of the induced current?



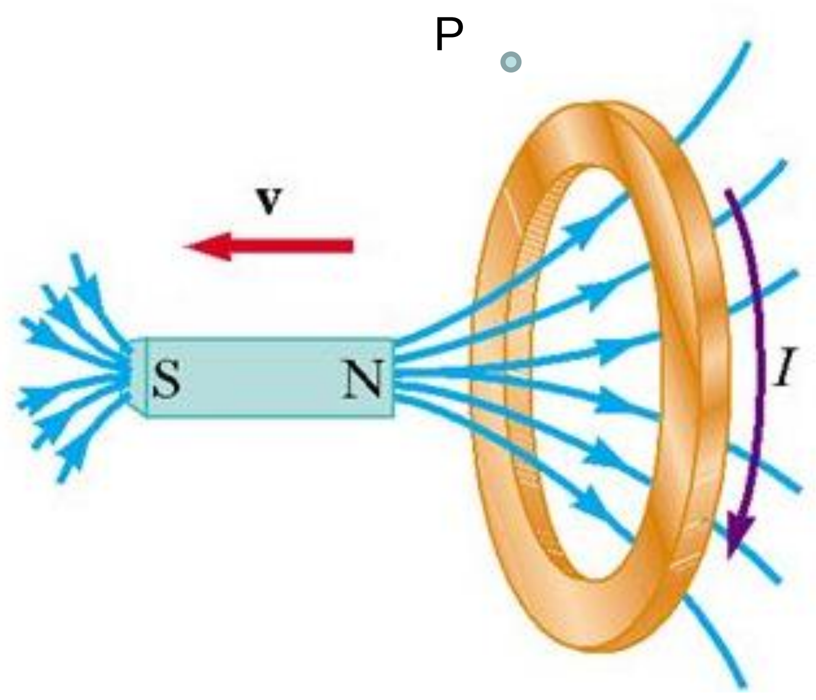




Please type: C (Clockwise) or CC (counterclockwise)

Retrieval Session

What is the electric field in point P ?



Please type: zero or Not zero

Static vs Time-Varying Electric Field

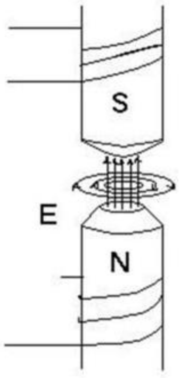
Static electric field

- Produced by static electric charges
- Electric field lines start on the positive charge(s) and end on negative charge(s)
- The work done moving charge on a closed contour is zero

$$\oint_L \vec{E}\,d\vec{l} = \oint_L E\,dl\cos\alpha = \oint_L E_l\,dl \equiv o$$

Time-varying electric field

- Produced by time-varying magnetic field
- Electric field lines are closed contours
- The work done moving charge on a closed contour may be NOT zero

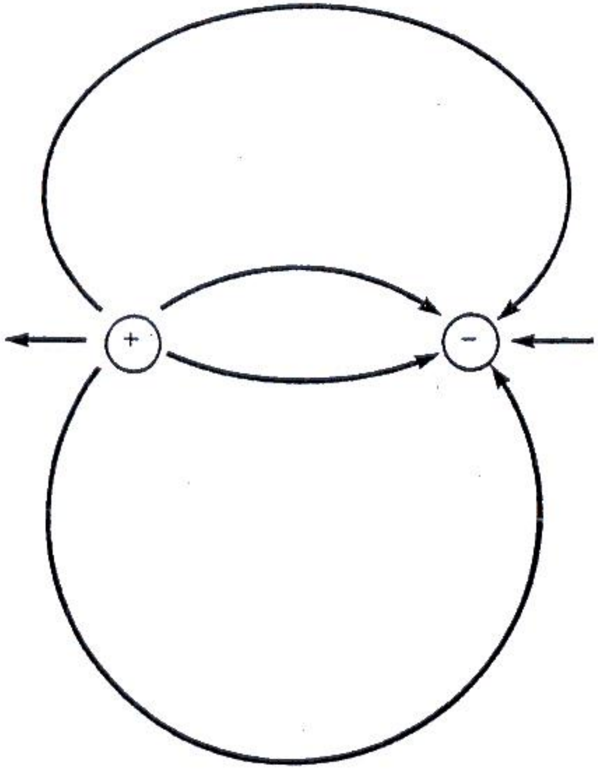


$$\oint_c \mathbf{E}\cdot d\ell = -\frac{d}{dt}\int_s \mathbf{B}\cdot d\mathbf{s}$$

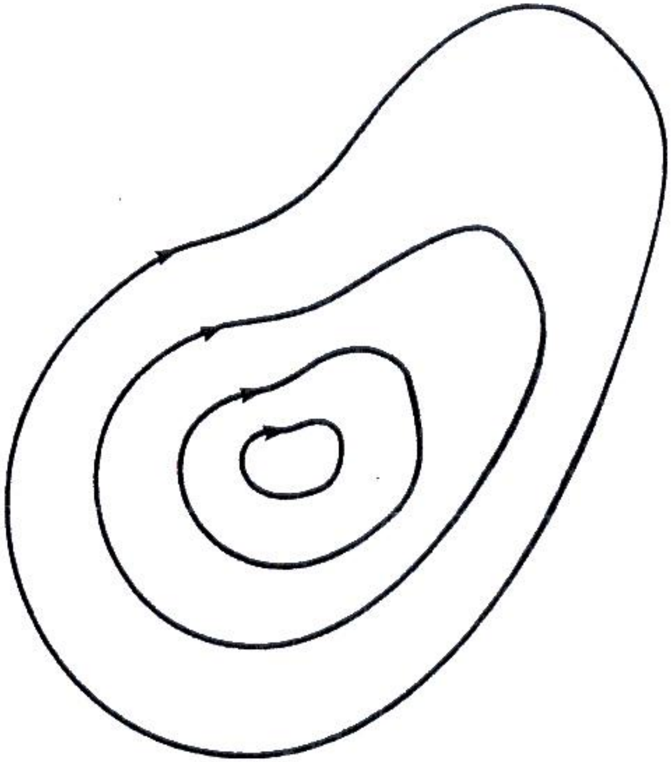
Different forms of Electric Field

Which of the graphs (left or right) shows electric field?

Electric field produced by charges



Vortex electric field (e.g. produced by time varying magnetic field)

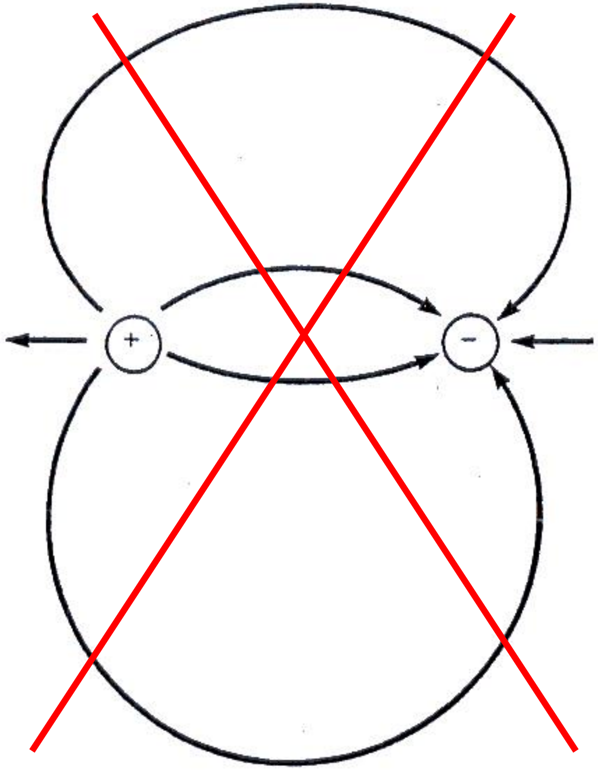


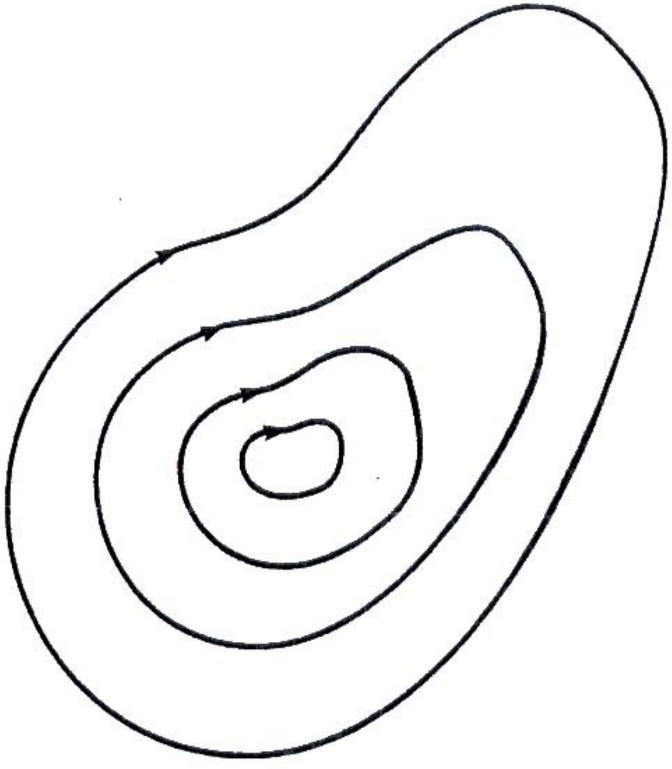
Different forms of Magnetic Field

Which of the graphs (left or right) shows magnetic field?

There are no magnetic charges!

Vortex magnetic field (e.g. produced by time varying electric field)





Lorentz Force

23. A charged particle (charge q, mass m) is moving under the influence of a magnetic field

$$\mathbf{B} = B_o \, \mathbf{a}_z .$$

If the velocity vector of the particle is

$$\mathbf{v} = v_x \, \mathbf{a}_x + v_y \, \mathbf{a}_y + v_z \, \mathbf{a}_z$$

Show that the acceleration of this particle lies in the x-y plane.

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

$\mathbf{F}$ = force

q = electric charge

$\mathbf{E}$ = external electric field

$\mathbf{v}$ = velocity

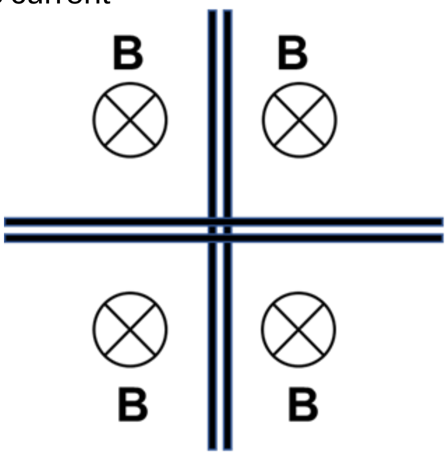
$\mathbf{B}$ = magnetic field

collegesearch

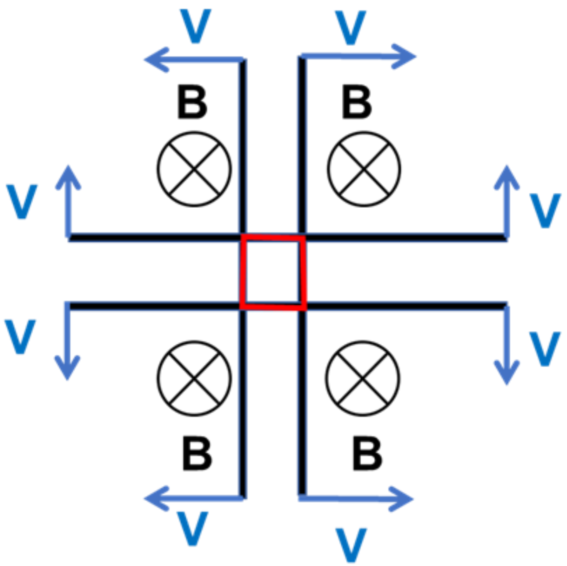
1. (**5 points**) A charged particle (charge q, mass m) is moving under the influence of a magnetic field **B**=$B_o\boldsymbol{a_x}$. If the velocity vector of the particle is **v**=$v_x\boldsymbol{a_x}+v_y\boldsymbol{a_y}+v_z\boldsymbol{a_z}$, show that the acceleration of this particle lies in the y-z plane.

$$\vec{v}\times\vec{B} = -B_0 v_y \overrightarrow{a_z} + B_0 v_z \overrightarrow{a_y}$$

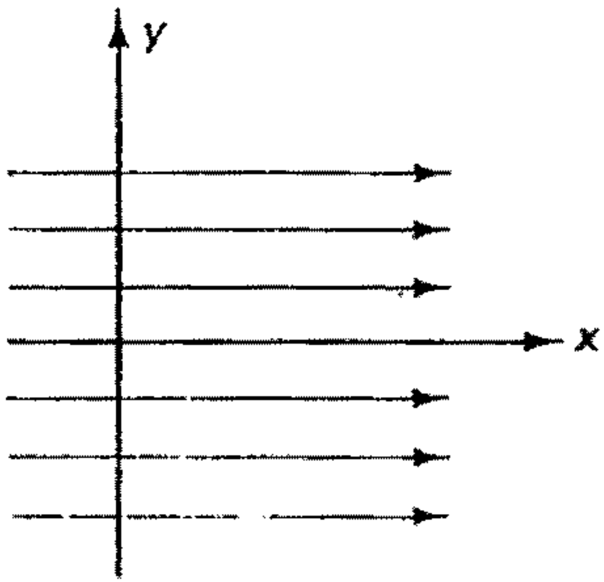
6. (**30 points**) The graph below shows conducting contours in a constant magnetic field B = 0.5T. All four sides of the contours move with constant speed v =0.1 m/s away from each other. The resistance per length is 0.1 Ω /m.
a) Calculate the induced electric current in both contours at t=2s
a) Show the direction of the induced electric current
•**counterclockwise**



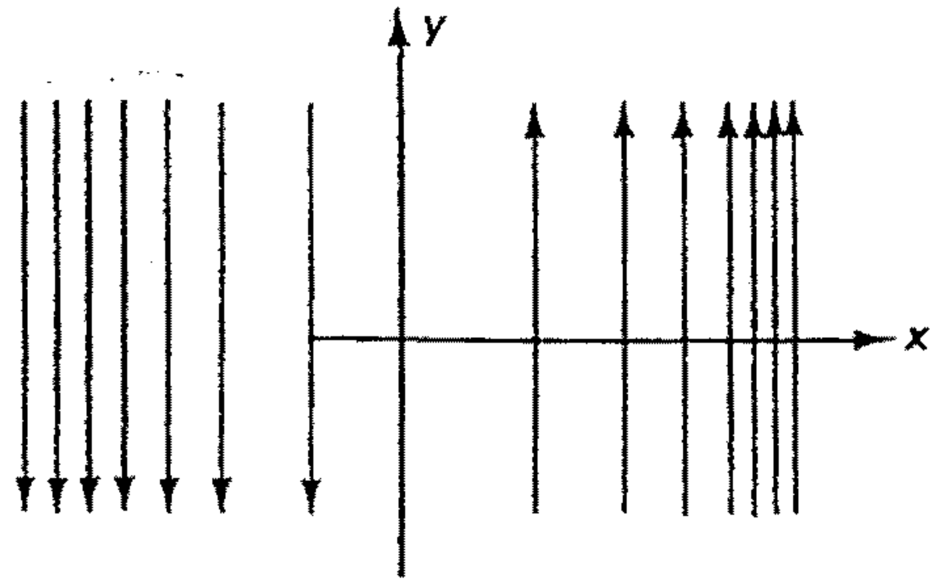
Starting moment t=0



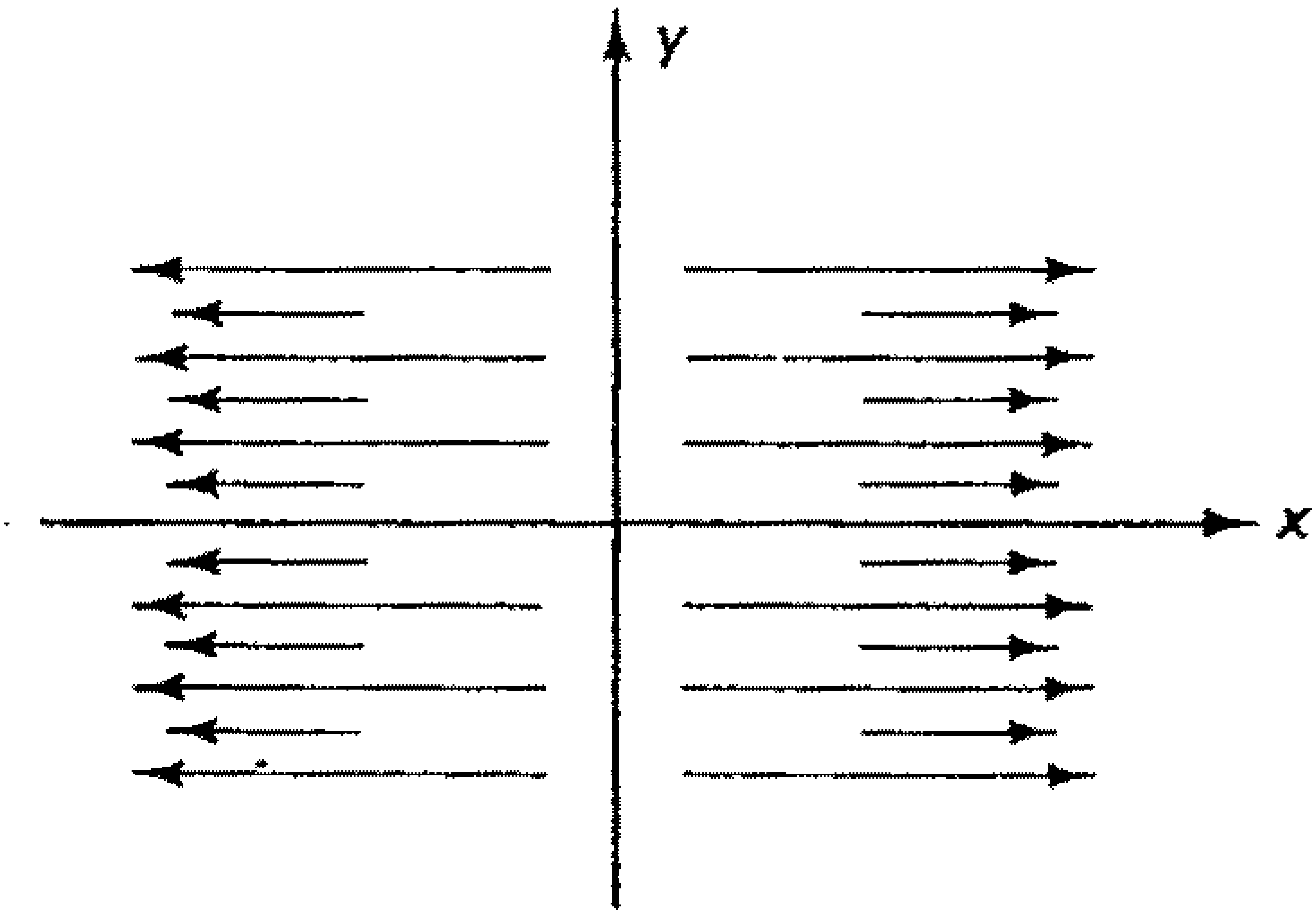
After some time t



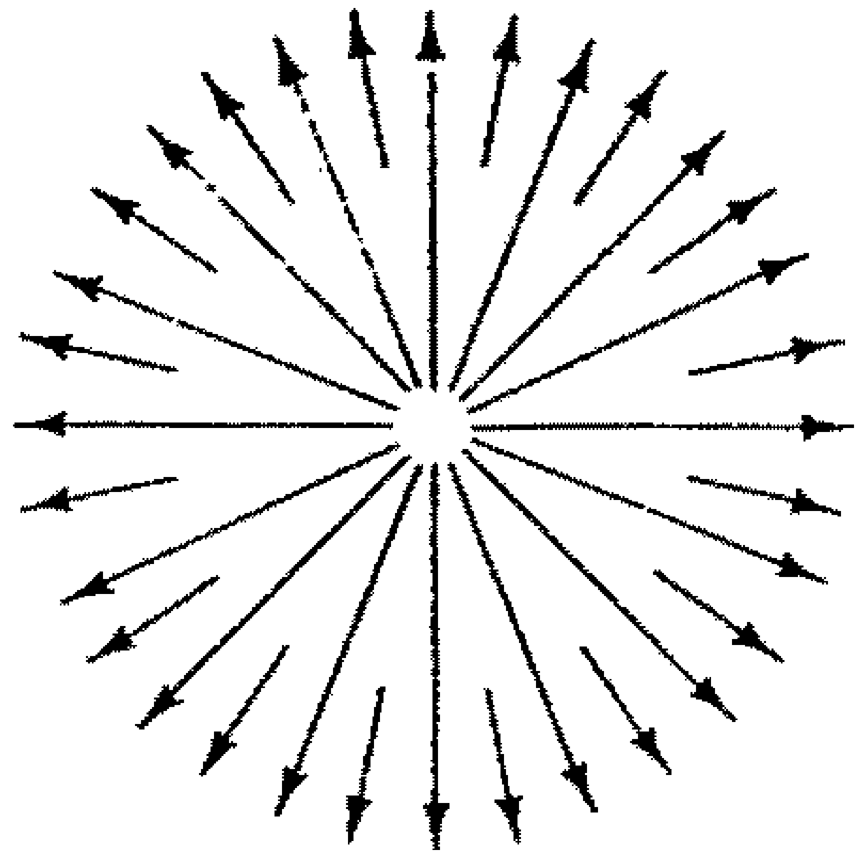
(a) $\mathbf{A} = K\,\mathbf{a}_x$



(b) $\mathbf{B} = Kx\,\mathbf{a}_y$



(c) $\mathbf{C} = Kx\,\mathbf{a}_x$



(e) $\mathbf{E} = K\,\mathbf{a}_\rho$

Figure 1.27 Flux representation of various vectors.

Maxwell's Equations

INTEGRAL FORMS	DIFFERENTIAL FORMS	
$\oint_s \epsilon_o \epsilon_r \mathbf{E} \cdot d\mathbf{s} = \int_v \rho_v \, dv$	$\nabla \cdot \epsilon_o \epsilon_r \mathbf{E} = \rho_v$	(4.1)
$\oint_s \mathbf{B} \cdot d\mathbf{s} = 0$	$\nabla \cdot \mathbf{B} = 0$	(4.2)
$\oint_c \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt}\int_s \mathbf{B} \cdot d\mathbf{s}$	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	(4.3)
$\oint_c \mathbf{H} \cdot d\boldsymbol{\ell} = \int_s \mathbf{J} \cdot d\mathbf{s} + \boxed{\frac{d}{dt}\int_s \epsilon_o \epsilon_r \mathbf{E} \cdot d\mathbf{s}}$	$\nabla \times \mathbf{H} = \mathbf{J} + \boxed{\frac{\partial \epsilon_o \epsilon_r \mathbf{E}}{\partial t}}$	(4.4)

Maxwell introduced

Waves in free space

$$\vec{\nabla}\times\vec{\nabla}\times\vec{E} = -\frac{\partial}{\partial t}\vec{\nabla}\times\vec{B} \qquad (\text{3rd eq.})$$

$\vec{\nabla}\times\vec{B}$ from 4th eq: $\vec{\nabla}\times\vec{B} = \mu_0\epsilon_0\frac{\partial\vec{E}}{\partial t}$

$$\Rightarrow \vec{\nabla}\times\vec{\nabla}\times\vec{E} = -\frac{\partial}{\partial t}\left(\mu_0\epsilon_0\frac{\partial\vec{E}}{\partial t}\right) = -\mu_0\epsilon_0\frac{\partial^2\vec{E}}{\partial t^2}$$

Vector Identity: $\vec{\nabla}\times\vec{\nabla}\times\vec{E} = \vec{\nabla}(\vec{\nabla}\cdot\vec{E}) - \nabla^2\vec{E}$

$$\vec{\nabla}^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} \quad \text{– Laplacian operator}$$

$$\vec{\nabla}^2 E_x = \frac{\partial^2 E_x}{\partial x^2} + \frac{\partial^2 E_x}{\partial y^2} + \frac{\partial^2 E_x}{\partial z^2} \ldots$$

from 1st eq: $\vec{\nabla}\cdot\vec{E} = 0 \quad \Rightarrow \vec{\nabla}\times\vec{\nabla}\times\vec{E} = -\nabla^2\vec{E}$

$$-\nabla^2\vec{E} = -\mu_0\epsilon_0\frac{\partial^2\vec{E}}{\partial t^2}$$

$$\boxed{\nabla^2\vec{E} - \mu_0\epsilon_0\frac{\partial^2\vec{E}}{\partial t^2}} \qquad \boxed{\nabla^2\vec{B} - \mu_0\epsilon_0\frac{\partial^2\vec{B}}{\partial t^2} = 0}$$

– homogeneous vector wave equation

Maxwell's Discovery The Speed of Light

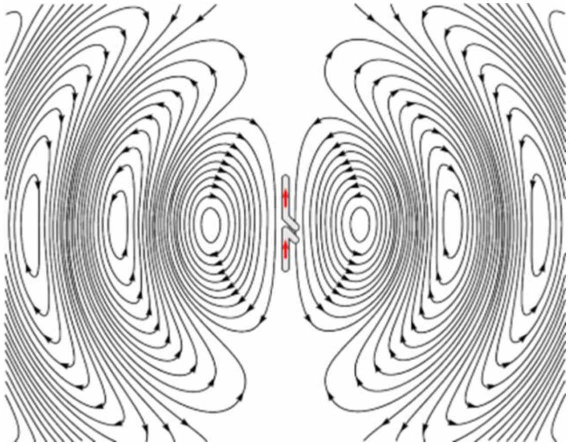
$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}$$

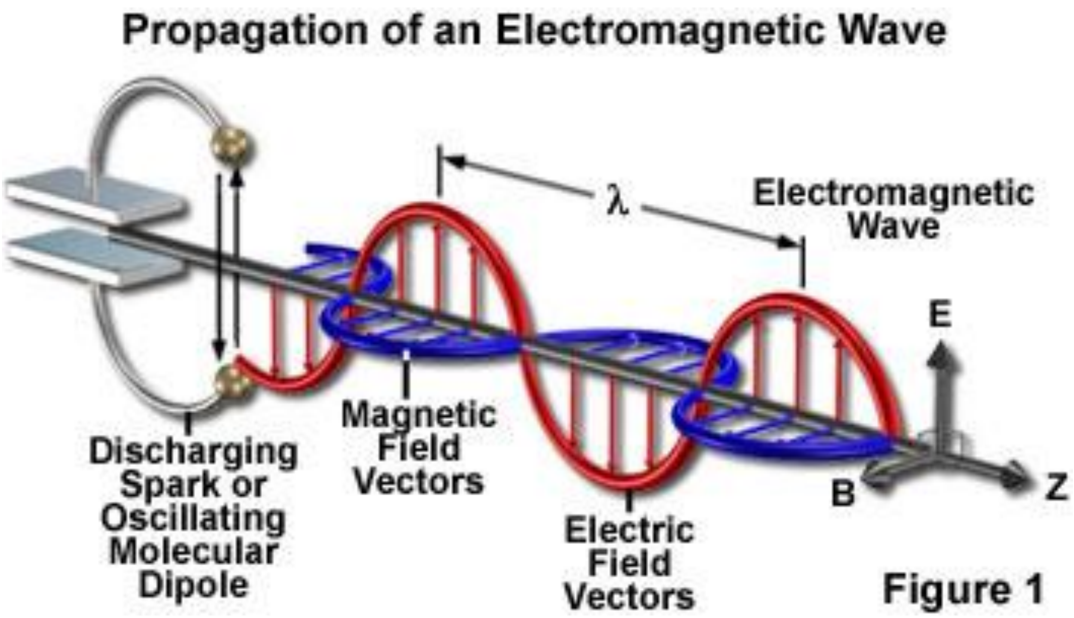
The speed of light was determined by many scientists using many different methods but this determination by Maxwell was serendipitous!

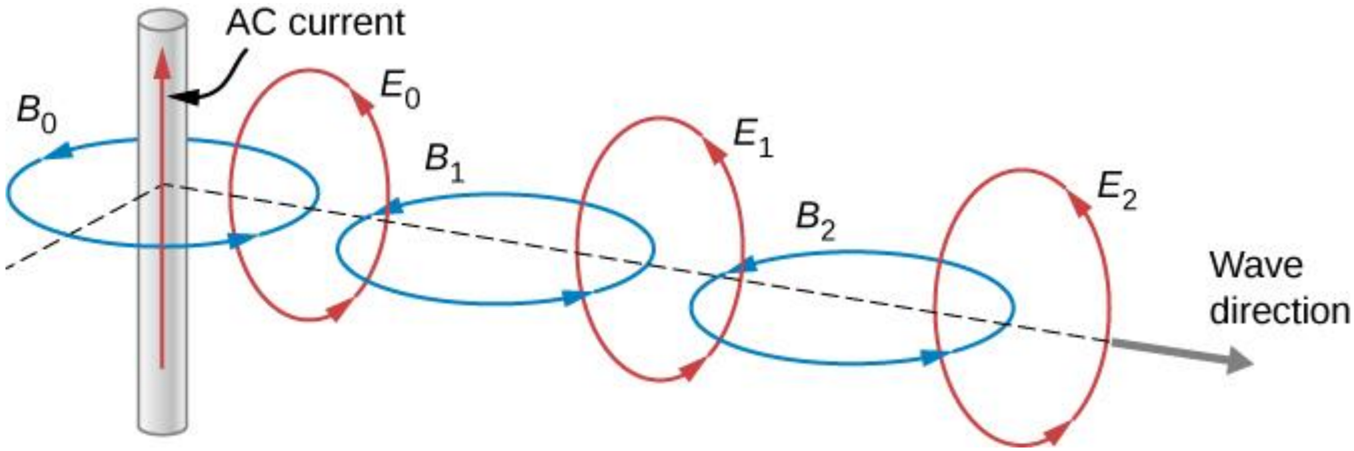
$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}} = \frac{1}{\sqrt{(8.85x10^{-12}\,\frac{C}{Nm^2})(4\pi x10^{-7}\,\frac{N}{A^2})}} = 3.00x10^8\,{}^{m}\!/_{s}$$

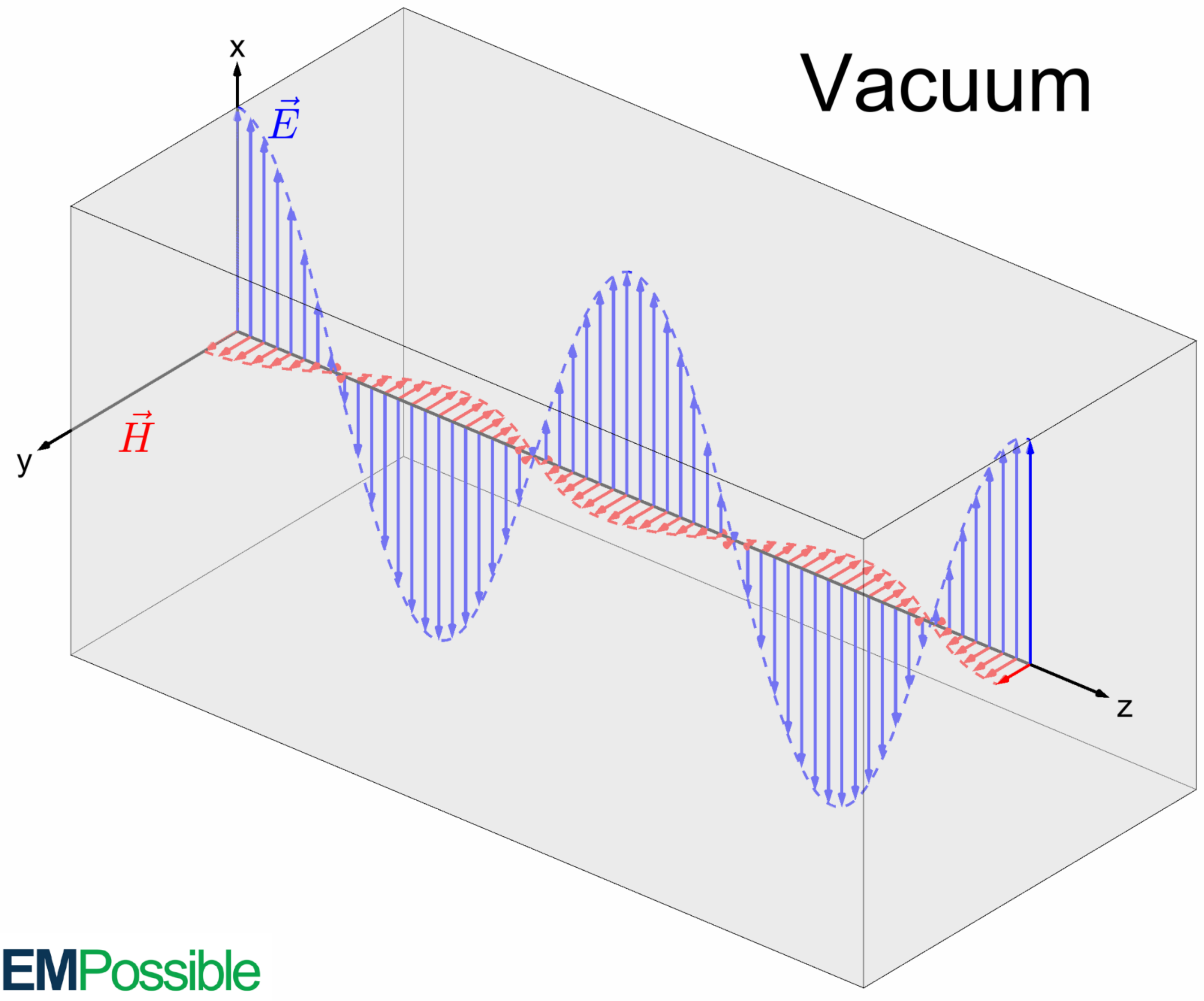
experimentally determined constants

Propagating Electromagnetic Wave



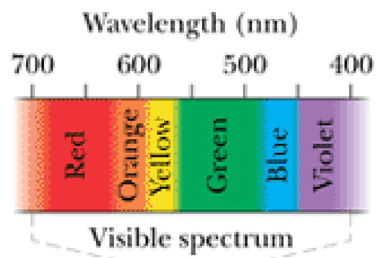


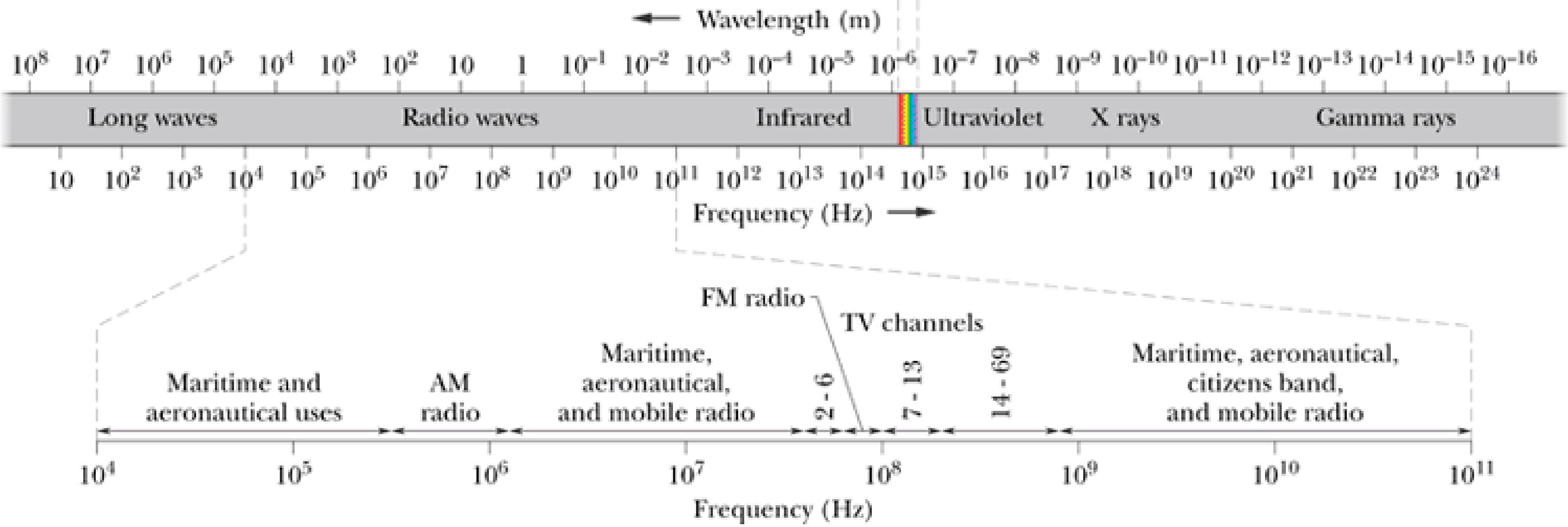


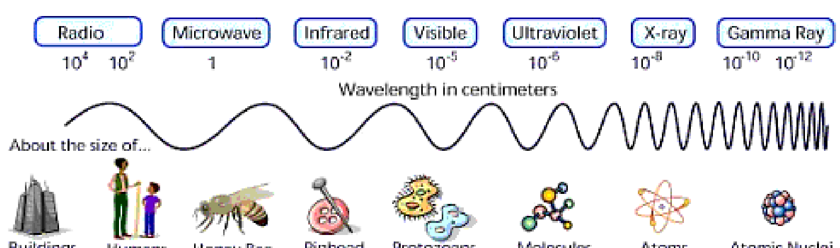

Vacuum
x
y
z
$\vec{E}$
$\vec{H}$

Maxwell's Rainbow

Extremely low frequency (ELF) is the ITU designation for electromagnetic radiation (radio waves) with frequencies from 3 to 30 Hz, and corresponding wavelengths of 100,000 to 10,000 kilometers, respectively). ELF waves can penetrate seawater, which makes them useful in communication with submarines. The US, Russia, India, and China are the only nations known to have constructed ELF communication facilities.

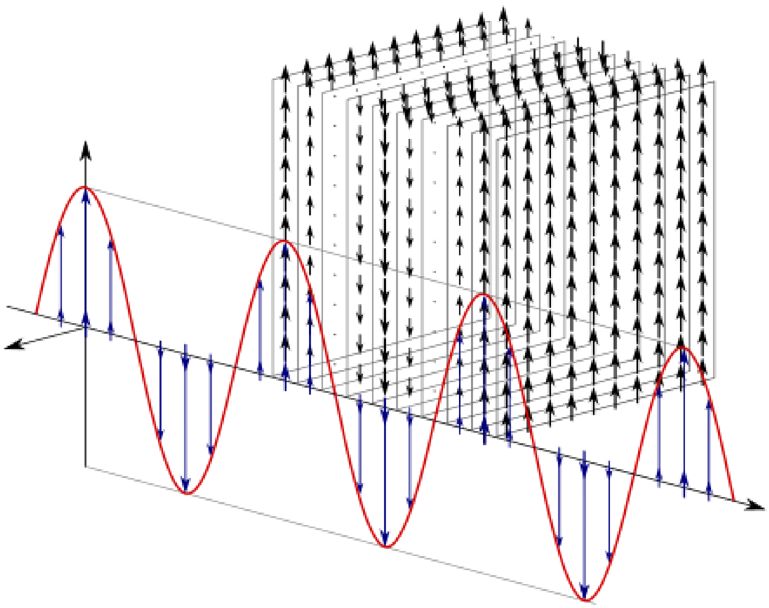




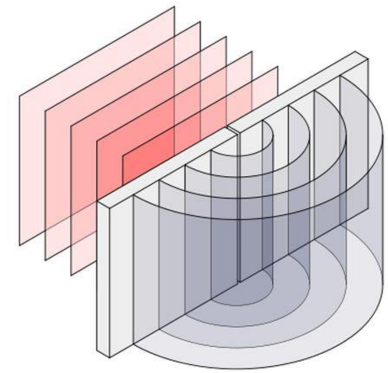


Plane, Cylindrical, and Spherical Electromagnetic Waves

Plane wave

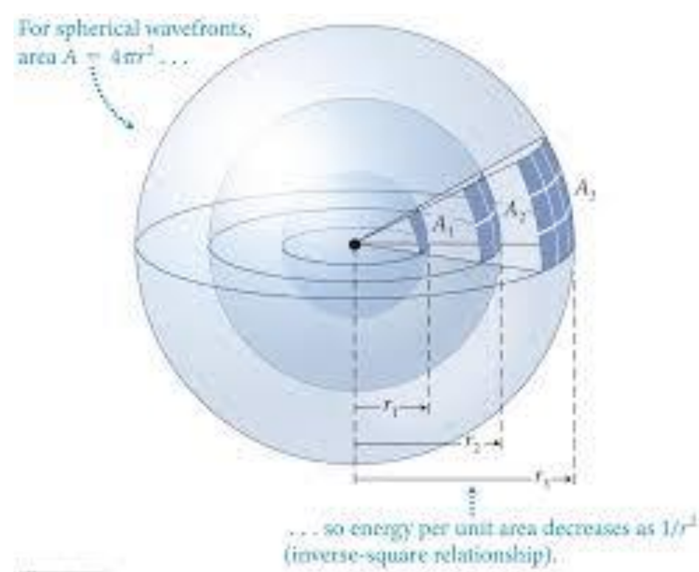


Cylindrical waves



$$\Psi = \left(\frac{A}{\sqrt{\rho}}\right)e^{i(k\rho \pm \omega t)}$$

Spherical Wave



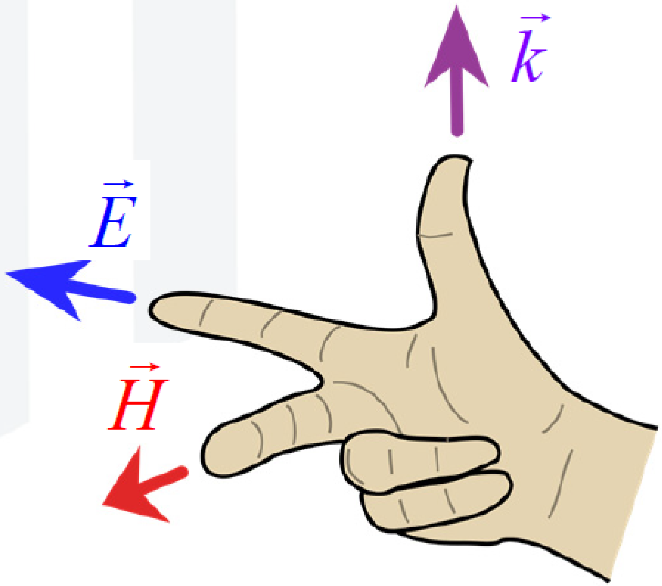
In physics, a wavefront is the locus of points characterized by propagation of positions of identical phase: propagation of a point in 1D, a curve in 2D or a surface in 3D. For an electromagnetic wave, the wavefront is represented as a surface of identical phase, and can be modified with conventional optics. For instance, a lens can change the shape of optical wavefronts from planar to spherical as the lens introduces a spatial phase variation across the beam shape.

Plane Waves in LHI Media are TEM

In linear, homogeneous, and isotropic (LHI) media, the electric field $\vec{E}$, magnetic field $\vec{H}$, and direction of the wave $\vec{k}$ are all perpendicular to each other. These are called transverse electromagnetic (TEM) waves.

$$\vec{E} \perp \vec{k} \perp \vec{H}$$

Further $\vec{E}$, $\vec{H}$, and $\vec{k}$ follow a right-hand rule.



Electro-magnetic waves
time-domain (instantaneous) expression

$$\vec{E}(z,t) = E_{max} \cos(\underbrace{2\pi \times ft}_{\omega} - \underbrace{\beta_0}_{\frac{2\pi}{\lambda}} z + \phi_0)\, \overrightarrow{a_x} \quad \text{V/m}$$

$$\vec{H}(z,t) = H_{max} \cos(2\pi \times ft - \beta_0 z + \phi_0)\, \overrightarrow{a_y} \quad \text{A/m}$$

- By defining

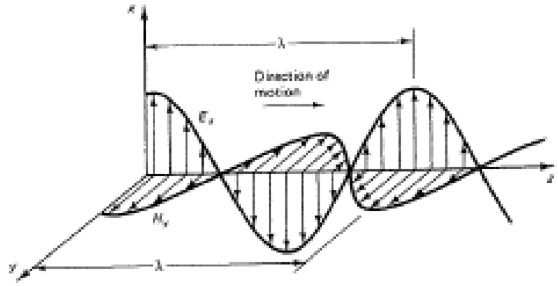
$$\text{radian frequency} = \omega = 2\pi f \quad \text{(rad/s)}$$

$$\text{wavenumber} = \beta = \frac{2\pi}{\lambda} \quad \text{(rad/m)}$$

Note: Radian frequency is also denoted *angular velocity* and wavenumber is also denoted *phase constant*

Propagating Waves

1



$$\hat{E}_x(z,t) = \hat{E}_m e^{-j\beta_0 z}$$

$$E_x(z,t) = E_m \cos(\omega t - \beta_0 z)$$

$$\hat{B}_y = \frac{\hat{E}_x}{c}$$

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}} \approx 3 \cdot 10^8 \, m/s$$

$\vec{E}$ and $\vec{B}$ are in phase

Technical/Engineering applications:

E [V/m]

$\hat{B}_M = \mu_0 \hat{H}_Y$, $\hat{H}$ - magnetic field intensity

H [A/m] $\Rightarrow$ $\dfrac{E_x}{H_y} = \mu_0 c = \sqrt{\dfrac{\mu_0}{\varepsilon_0}} = \eta_0 = 120\pi \approx$

$\boxed{\text{MKS units}}$ $\approx 377\,\Omega$ - intrinsic wave impedance

Wavelengh - distance z that the wave must travel to change the phase by 2π radians

$$\beta_0 \lambda = 2\pi \Rightarrow \boxed{\lambda = \frac{2\pi}{\beta_0} = \frac{c}{f}}$$

8. (**30 points**) The electric field of a uniform plane wave propagating in a dielectric nonconducting medium having $\mu = \mu_0$ is given by

$$\vec{E}(z,t) = 5\cos(4\pi \times 10^7 t - 0.2\pi z)\overrightarrow{a_x}, \qquad V/m$$

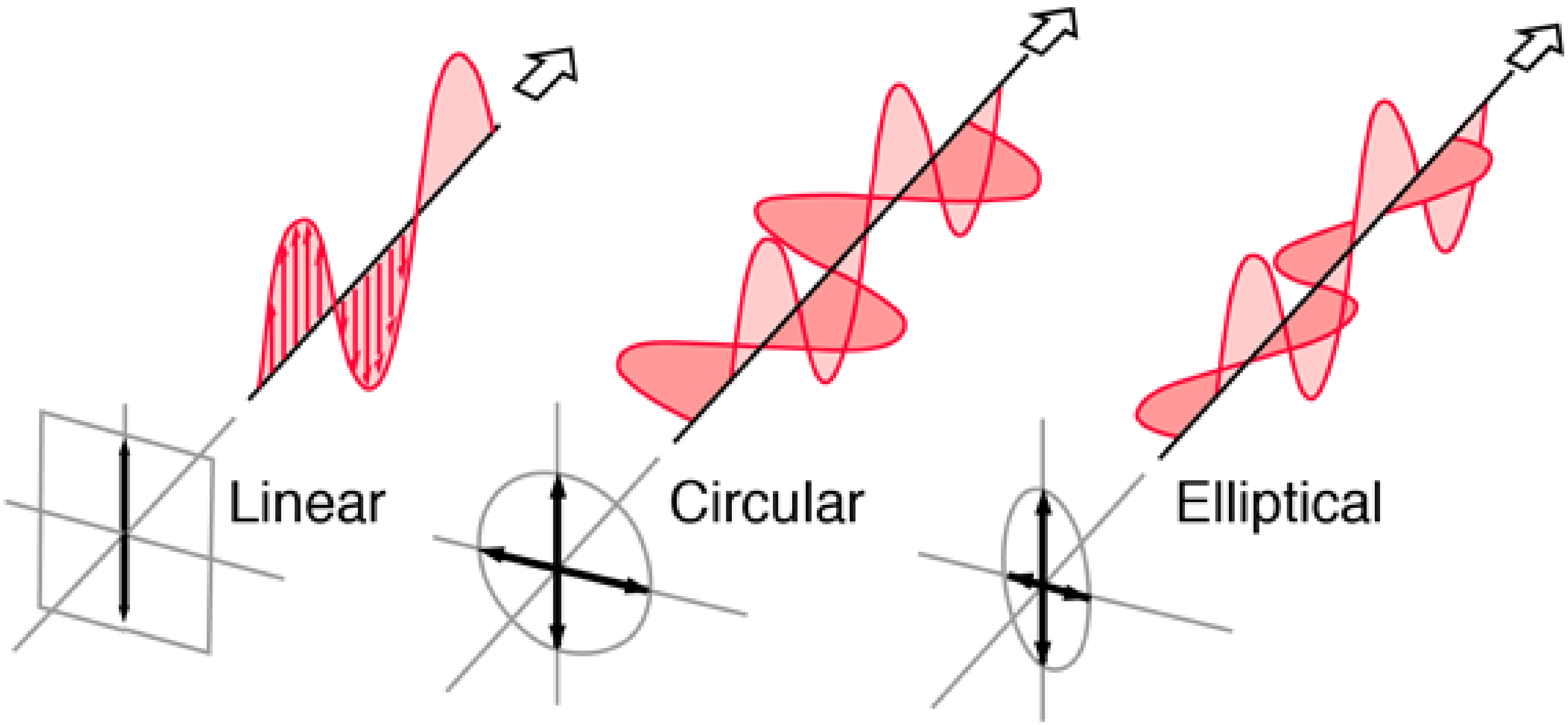
a) Frequency: **20 MHz**
b) Wavelength: **10 m**
c) Phase velocity: 2 **10^8 m/s**
d) Permittivity of medium $\epsilon = 2.25\,\epsilon_0$
e) Associated magnetic field vector **H**

$$\vec{H}(z,t) = 0.02\cos(4\pi \times 10^7 t - 0.2\pi z)\overrightarrow{a_y}, \qquad A/m$$

f) Poynting vector **P:**

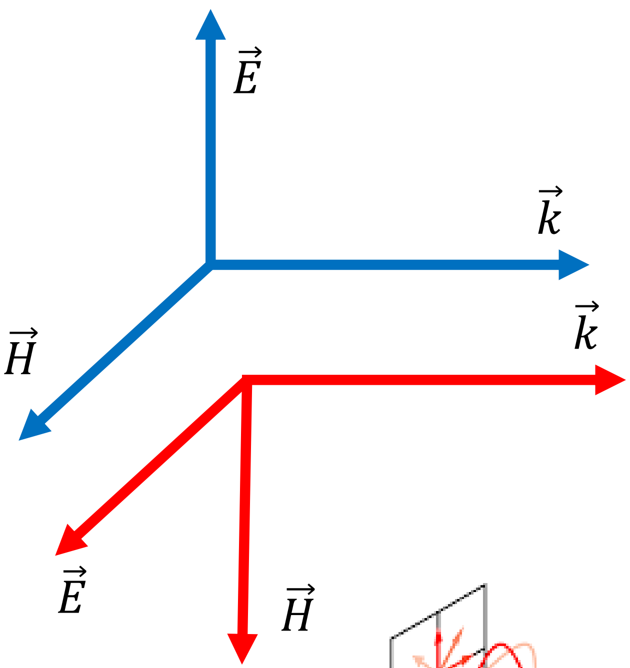
$$\vec{P} = 0.1\cos\text{^}2(4\pi \times 10^7 t - 0.2\pi z)\overrightarrow{a_z}, \qquad W/m^2$$

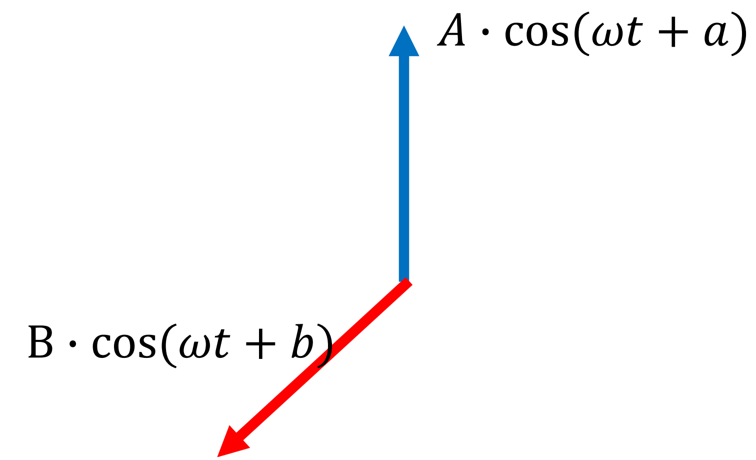
Polarization of Plane Wave



EM Polarization

The polarization of an electromagnetic wave describes the orientation of the oscillating electric field.

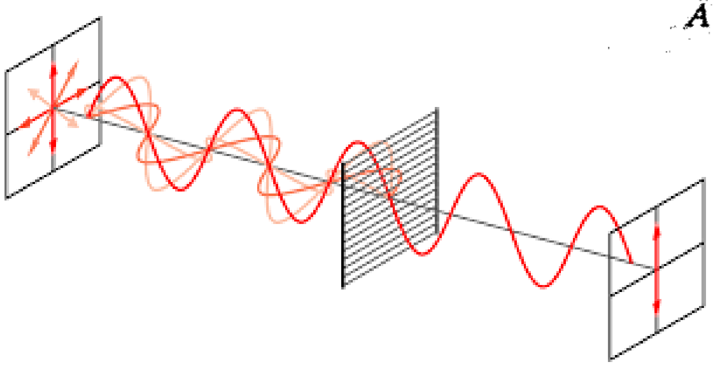




$$\hat{\mathbf{E}} = (\hat{A}\,\mathbf{a}_x + \hat{B}\,\mathbf{a}_y)e^{-j\beta z}$$

where the amplitudes $\hat{A}$ and $\hat{B}$ may be complex, that is,

$$\hat{A} = |\hat{A}|e^{ja}, \qquad \hat{B} = |\hat{B}|e^{jb}$$



Polarization of Plane Wave

$$\hat{\mathbf{E}} = (\hat{A}\,\mathbf{a}_x + \hat{B}\,\mathbf{a}_y)e^{-j\beta z}$$

where the amplitudes $\hat{A}$ and $\hat{B}$ may be complex, that is,

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Elliptical

$a \neq b;\ |A| \neq 0;\ |B| \neq 0$

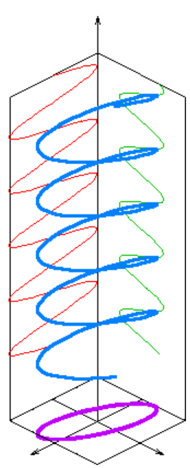
$|A| \neq |B|$

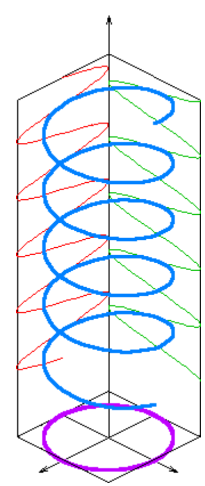
Circular

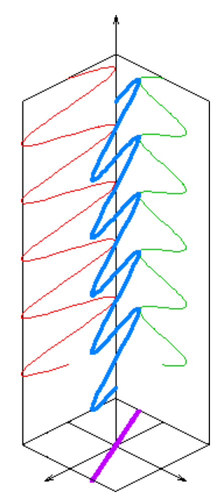
$|A| = |B|\ and\ |a - b| = \pi/2$

Linear

$a = b$







Retrieval Session

Electric field of EM wave is given by:

$\hat{E} = \widehat{E_x} + \widehat{E_y};$

$\widehat{E_x} = 4\ (V/m) \cdot e^{j\pi/7}$

$\widehat{E_y} = 5\ (V/m) \cdot e^{j\pi/7}$

The wave polarizations:

a) Linear
b) Circular
c) Elliptical
d) Not possible to define

Retrieval Session

Electric field of EM wave is given by:

$$\hat{E} = \widehat{E_x} + \widehat{E_y};$$

$$\widehat{E_x} = 3\ (V/m) \cdot e^{j\pi/7}$$

$$\widehat{E_y} = 2\ (V/m) \cdot e^{j\pi/2}$$

The wave polarizations:

a) Linear

b) Circular

c) Elliptical

d) Not possible to define

Retrieval Session

Electric field of EM wave is given by:

$\hat{E} = \widehat{E_x} + \widehat{E_y};$

$\widehat{E_x} = 7\ (V/m) \cdot e^{j\pi}$

$\widehat{E_y} = 7\ (V/m) \cdot e^{j2\pi}$

The wave polarizations:

a) Linear
b) Circular
c) Elliptical
d) Not possible to define

Retrieval Session

Electric field of EM wave is given by:

$$\hat{E} = \widehat{E_x} + \widehat{E_y};$$

$$\widehat{E_x} = 3\ (V/m) \cdot e^{j\pi}$$

$$\widehat{E_y} = -3\ (V/m) \cdot e^{j\pi/2}$$

The wave polarizations:

a) Linear
b) Circular
c) Elliptical
d) Not possible to define

Divergence of a vector $\vec{F}$ is a measure of the outflow of flux from a small closed surface per unit volume V as $V \to 0$

$$\boxed{\text{Divergence of } \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}}$$

$$\boxed{\text{div } \vec{F} \equiv \vec{\nabla} \cdot \vec{F}}$$

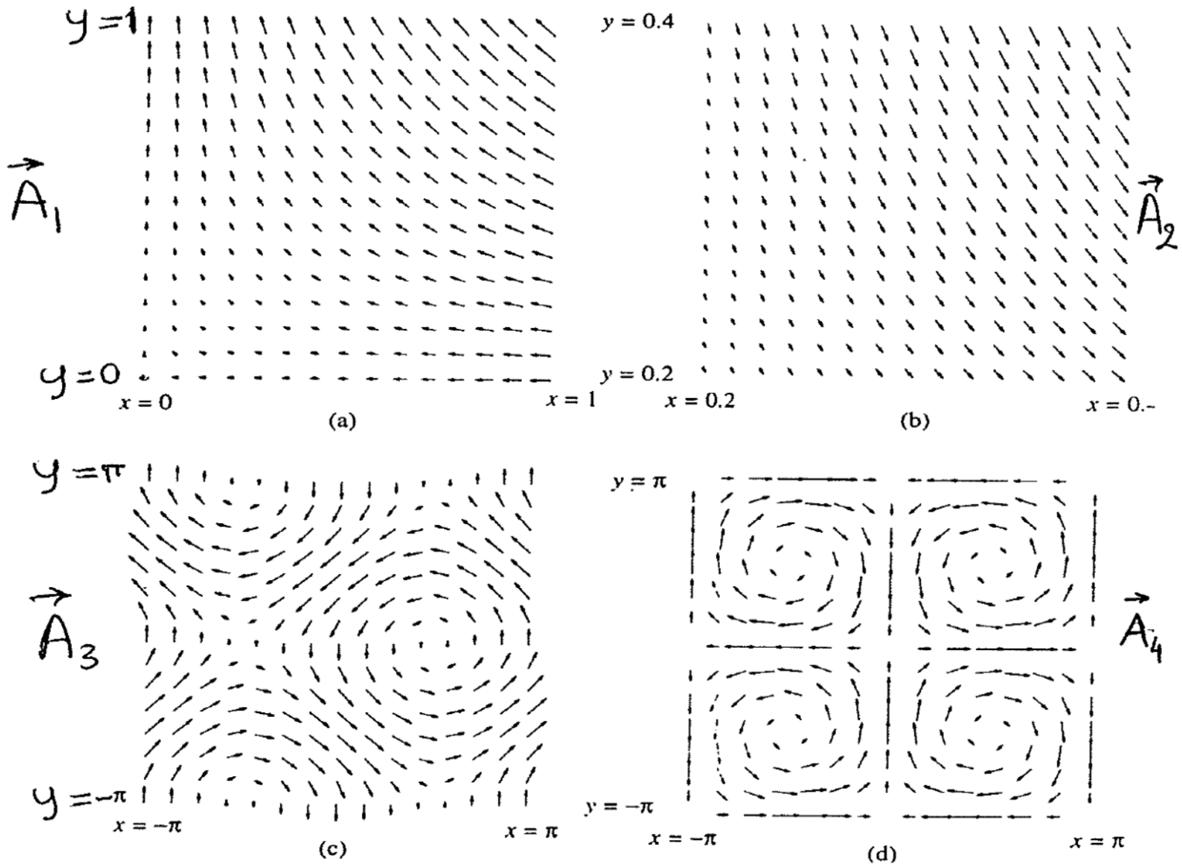
$$\vec{\nabla} \cdot \vec{F} = \left(\vec{a}_x \frac{\partial}{\partial x} + \vec{a}_y \frac{\partial}{\partial y} + \vec{a}_z \frac{\partial}{\partial z}\right)\left(F_x \vec{a}_x + F_y \vec{a}_y + F_z \vec{a}_z\right)$$

$$= \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

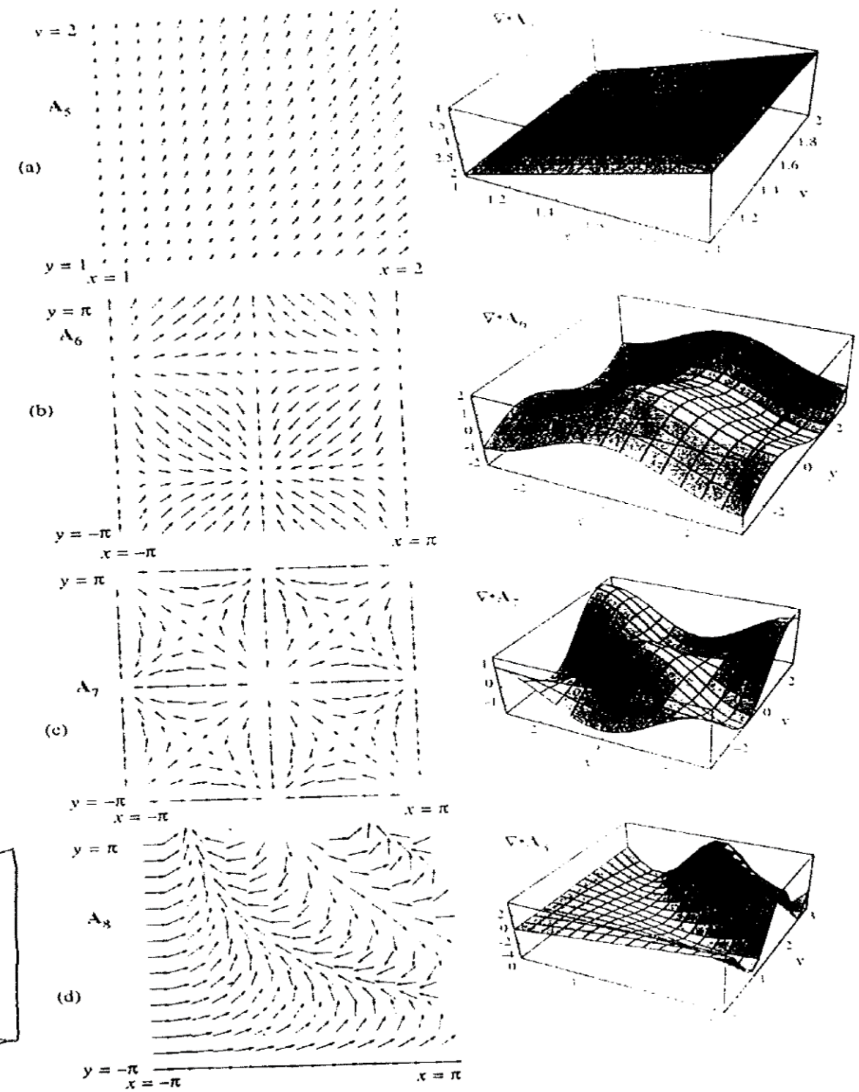
$$\text{div } \vec{F} = \frac{1}{\rho} \frac{\partial (F_\rho \rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z} \quad \text{(cylind.)}$$

$$\text{div } \vec{F} = \frac{1}{r^2} \frac{\partial (F_r r^2)}{\partial r} + \frac{1}{r \sin\theta} \frac{\partial (F_\theta \sin\theta)}{\partial \theta} + \frac{1}{r \sin\theta} \frac{\partial F_\phi}{\partial \phi} \quad \text{(spherical)}$$

Divergence - Free Fields

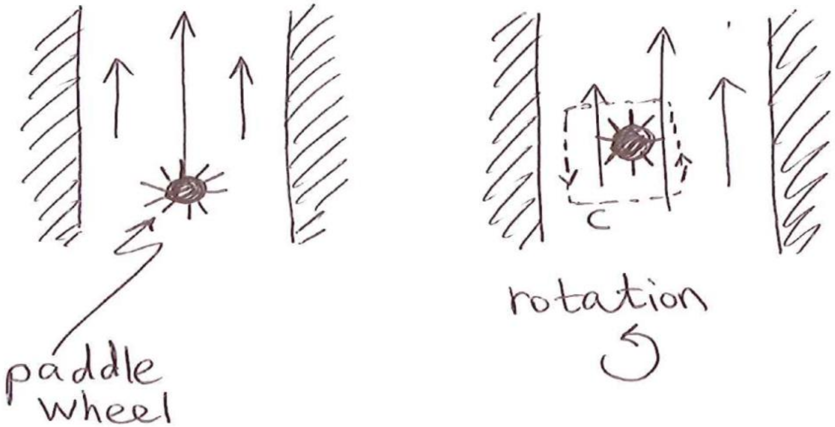


Vector Fields with Non-Zero Divergence



$$\boxed{\vec{\nabla} \cdot \epsilon_0 \vec{E} = \rho_v}$$

Curl of Vector Field ②



$$\Rightarrow \oint_C \vec{F}\cdot d\vec{\ell} \neq 0$$

$$\mathrm{curl}\,\vec{F} = [\mathrm{curl}\,\vec{F}]_x\,\vec{a}_x + [\mathrm{curl}\,\vec{F}]_y\,\vec{a}_y + + [\mathrm{curl}\,\vec{F}]_z\,\vec{a}_z$$

$$\boxed{[\mathrm{curl}\,F]_x = \lim_{\substack{\Delta y \to 0 \\ \Delta z \to 0}} \frac{\oint_C \vec{F}\cdot d\vec{\ell}}{\Delta y\,\Delta z}}$$

$$\mathrm{curl}\,\vec{F} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = \vec{\nabla}\times\vec{F}$$

"The curl of a vector is a point form expression of the line integral of a vector field around a closed contour"

For Cartesian coordinate:

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$\nabla \times \mathbf{A} = \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] \mathbf{a}_x - \left[\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z} \right] \mathbf{a}_y + \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right] \mathbf{a}_z$$

For Circular cylindrical coordinate:

$$\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \mathbf{a}_\rho & \rho \mathbf{a}_\phi & \mathbf{a}_z \\ \dfrac{\partial}{\partial \rho} & \dfrac{\partial}{\partial \phi} & \dfrac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

$$\nabla \times \mathbf{A} = \left[\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right] \mathbf{a}_\rho - \left[\frac{\partial A_z}{\partial \rho} - \frac{\partial A_\rho}{\partial z} \right] \mathbf{a}_\phi$$

$$+ \frac{1}{\rho} \left[\frac{\partial (\rho A_\phi)}{\partial \rho} - \frac{\partial A_\rho}{\partial \phi} \right] \mathbf{a}_z$$

For Spherical coordinate:

$$\nabla \times \mathbf{A} = \frac{1}{r^2 \sin\theta} \begin{vmatrix} \mathbf{a}_r & \mathbf{a}_\theta & \mathbf{a}_\phi \\ \dfrac{\partial}{\partial r} & \dfrac{\partial}{\partial \theta} & \dfrac{\partial}{\partial \phi} \\ A_r & rA_\theta & (r\sin\theta)A_\phi \end{vmatrix}$$

$$\nabla \times \mathbf{A} = \frac{1}{r\sin\theta}\left(\frac{\partial(\sin\theta A_\phi)}{\partial\theta} - \frac{\partial A_\theta}{\partial\phi}\right)\mathbf{a}_r + \frac{1}{r}\left(\frac{1}{\sin\theta}\frac{\partial A_r}{\partial\phi} - \frac{\partial(rA_\phi)}{\partial r}\right)\mathbf{a}_\theta$$

$$+ \frac{1}{r}\left(\frac{\partial(rA_\theta)}{\partial r} - \frac{\partial(A_r)}{\partial\theta}\right)\mathbf{a}_\phi$$

EXAMPLE 2.21

Determine which of the following vectors may represent a static magnetic field. If so, calculate the current density associated with it.

1. $\mathbf{B} = x\,\mathbf{a}_x - y\,\mathbf{a}_y$
2. $\mathbf{B} = \rho\,\mathbf{a}_\phi$
3. $\mathbf{B} = r\cos\phi\,\mathbf{a}_r - 3r\sin\theta\sin\phi\,\mathbf{a}_\phi$

Solution

For a vector field to represent a magnetic flux density vector, it should satisfy Gauss's law, hence,

1.
$$\nabla \cdot B = 0$$
$$\nabla \cdot \mathbf{B} = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z}$$
$$= 0$$

The **B** vector in part 1 may therefore represent a magnetic flux density vector. The current density associated with it is obtained using Ampere's law, which for static ($\partial/\partial t = 0$) fields reduces to

$$\nabla \times \frac{\mathbf{B}}{\mu_o} = \mathbf{J}$$

$$\therefore \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ x & -y & 0 \end{vmatrix} = \mu_o\,\mathbf{J}$$

Carrying out the curl analysis, we obtain

$$\mathbf{J} = 0$$

2. $$\nabla \cdot \mathbf{B} = \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho B_\rho) + \frac{1}{\rho}\frac{\partial B_\phi}{\partial \phi} + \frac{\partial B_z}{\partial z}$$
$$= 0$$

The vector **B** in part 2 may also represent magnetic flux density vector. The current density **J** is obtained from

$$\nabla \times \mathbf{B} = \mu_o \mathbf{J}$$
$$= \left[\frac{1}{\rho}\left(\frac{\partial B_z}{\partial \phi} - \frac{\partial B_\phi}{\partial z}\right)\right]\mathbf{a}_\rho + \left[\frac{\partial B_\rho}{\partial z} - \frac{\partial B_z}{\partial \rho}\right]\mathbf{a}_\phi + \left[\frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho B_\phi) - \frac{1}{\rho}\frac{\partial B_\rho}{\partial \phi}\right]\mathbf{a}_z$$

Substituting B_ρ, B_ϕ, and B_z of this vector, we obtain

$$\mathbf{J} = \frac{2}{\mu_o}\mathbf{a}_z$$

3. $$\nabla \cdot \mathbf{B} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 B_r) + \frac{1}{r \sin\theta}\frac{\partial}{\partial \theta}(B_\theta \sin\theta) + \frac{1}{r\sin\theta}\frac{\partial B_\phi}{\partial \phi}$$
$$= 3\cos\phi + \frac{1}{r\sin\theta}(-3r\sin\theta\cos\phi)$$
$$= 0$$

B may represent magnetic flux density vector.

$$\nabla \times \mathbf{B} = \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial \theta}(B_\phi \sin\theta) - \frac{\partial B_\theta}{\partial \phi}\right]\mathbf{a}_r$$
$$+ \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial B_r}{\partial \phi} - \frac{\partial}{\partial r}(rB_\phi)\right]\mathbf{a}_\theta + \frac{1}{r}\left[\frac{\partial}{\partial r}(rB_\theta) - \frac{\partial B_r}{\partial \theta}\right]\mathbf{a}_\phi$$
$$= \mu_o \mathbf{J}$$

Substituting B_r, B_θ, and B_ϕ components, we obtain

$$\mathbf{J} = \frac{1}{\mu_o}\left[-6\cos\theta\sin\phi\,\mathbf{a}_r + \left(6\sin\theta\sin\phi - \frac{\sin\phi}{\sin\theta}\right)\mathbf{a}_\theta\right]$$

Example 2.23

$$\vec{H}(z,t) = \underbrace{4\times10^{-6}}_{\hat{H}_y(z)} \cos(\underbrace{2\pi\times10^{7}}_{\omega} t - \beta_0 z)\,\vec{a_y} \quad [A/m]$$

$$\beta_0 = \frac{\omega}{c} = \frac{2\pi\times10^{7}}{3\times10^{8}} = 0.21 \text{ rad/m}$$

$$\frac{\hat{E}_x}{\hat{H}_y} = \eta_0 \quad \Rightarrow \quad \hat{E}_x = 4\times10^{-6}\cdot\eta_0$$

$$\vec{E}(z,t) = \eta_0 \times 4\times10^{-6} \cos(2\pi\times10^{7} t - \beta_0 z)\,\vec{a_x}$$

$$= 1.51\times10^{-3} \cos(2\pi\times10^{7} t - 0.21 z)\,\vec{a}_x$$

$$[V/m]$$

Example 2.24

$E_x = 100\ V/m$; $\lambda = 25\ cm$

$$c = f \cdot \lambda$$

$$f = \frac{3 \times 10^8\ m/s}{0.25\ m} = 12 \times 10^8\ Hz = 1.2\ GHz$$

$$\hat{E}(z) = 100\ e^{-j\beta_0 z}\ \vec{a}_x$$

$$\beta_0 = \frac{2\pi}{\lambda} = 8\pi/m$$

$$\hat{H}(z) = \frac{\hat{E}_z}{\eta_0} = \frac{100}{377}\ e^{-j8\pi z}\ \vec{a}_y$$

$$= 0.27\ e^{-j8\pi z}\ \vec{a}_y$$

$$\vec{E}(z,t) = 100\ \cos(24\pi \times 10^8 t - 8\pi z)\ \vec{a}_x$$

$$\vec{H}(z,t) = 0.27\ \cos(24\pi \times 10^8 t - 8\pi z)\ \vec{a}_y$$

Questions?